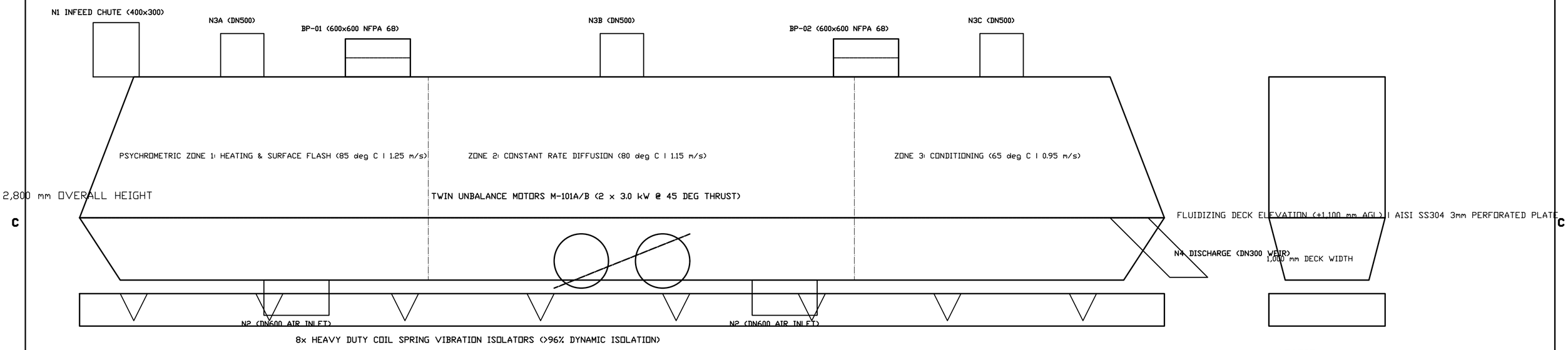


MARK	SERVICE	SIZE	RATING	FACING	LOCATION
N1	Moist Dal Infeed	400x300	FLANGED	FLAT FACE	Top Hood Infeed End
N2A/B	Hot Air Supply	2x DN600	150#	RAISED FACE	Bottom Plenum Underside
N3A-C	Exhaust Air Outlets	3x DN500	150#	FLAT FACE	Top Hood Crown
N4	Dry Dal Discharge	DN300	150#	RAISED FACE	Discharge End Retention Weir
N5	Plenum Fines Drain	DN100	150#	RAISED FACE	Plenum Invert Low Point
M1-M4	Inspection Doors	4x 500x400	HINGED	FOOD GASKET	Drive & Non-Drive Sides
BP-01/02	NFPA 68 Burst Panels	2x 600x600	SS COMPOSITE	ROOF DUCTED	Expansion Hood Crown

MECHANICAL, VIBRATION DYNAMICS & NFPA 68 SAFETY SPECIFICATION	
ENGINEERING PARAMETER	**VERIFIED SPECIFICATION & DESIGN CRITERIA**
Equipment Tag & Service	DR-101	Continuous Vibrating Fluid Bed Dal Conditioning & Drying
Effective Deck Dimensions	14,000 mm Length x 1,000 mm Width	Effective Deck Area: 14.0 m2
Material of Construction	Product Contact: AISI SS304 (Ra < 0.8 um sanitary polish)	Non-contact: IS2062 CS
Fluidizing Air Dynamics	14,150 kg/h (11,800 m3/h @ 85 deg C)	Superficial Velocity: 1.15 m/s (Umf = 0.85 m/s)
Thermal Duty & Water Evap	Water Evaporation: 412.34 kg/h H2O	Heat Duty: 512.4 kWth via Steam Coil HE-101
Vibration Dynamics & Motors	Stroke amplitude 3.5 mm @ 16 Hz (960 RPM); Twin unbalance motors 2 x 3.0 kW @ 45 deg
Spring Isolation Systen	8x Heavy-duty helical coil springs with rubber snubbers (> 96% dynamic isolation)
Residence Time & Control	18.0 – 22.0 minutes plug-flow with adjustable discharge overflow retention weir
NFPA 68 Explosion Relief	Dual 600x600 mm composite burst panels (Pstat = 0.10 bar g, Pred < 0.20 bar g)
Combustible Dust Rating	St-1 Agricultural Pulse Dust (Kst = 85 bar-m/s, Pmax = 7.8 bar g), exterior ducted

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.

15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GENERAL ARRANGEMENT - FLUID BED DRYER DR-101

DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-008	REV: V8.0	SHEET: 8 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ